Daily arXiv Research Briefing Agent

Group Report
Social Network Analysis, Spring 2026

Abstract

Daily arXiv Research Briefing Agent is a local Agent and Skills system for personalized paper discovery. Given a topic, optional arXiv category, date range, seed papers, feedback, or a follow-up query, the Agent retrieves candidate papers, ranks them with topic, seed, semantic, and feedback signals, and produces an evidence-aware daily briefing. The system integrates two independently testable public Skills. DiscoveryRecommendationSkill handles query planning, retrieval, recommendation, feedback refinement, and local-first follow-up filtering. ResearchSynthesisSkill handles structured paper extraction, briefing generation, and selected-paper explanation. A shared SQLite layer stores metadata, retrieval runs, seed preferences, feedback events, full-text cache rows, and embedding cache rows. Fixture-based evaluation gives precision@3, recall@3, MRR, rationale coverage, required-section coverage, and Top-K briefing coverage of 1.0. These results show reproducible workflow quality rather than large-scale benchmark superiority.

1 Functionality

Researchers face a daily filtering problem because arXiv publishes more papers than a user can inspect manually. This project turns a compact research interest into a ranked reading list, a structured daily briefing, and follow-up explanations with explicit evidence labels. The Agent treats topics, arXiv categories, papers, seed papers, user profiles, feedback events, and recommendation rationales as links in an academic information network. It therefore follows the course requirement for an Agent and Skills system: independent modules communicate through structured data, while the orchestrator composes them into one workflow.

The main entry point is DailyArxivAgentOrchestrator. It accepts requests from the command line interface or the Streamlit user interface and supports topic keywords, arXiv category, submitted-date range, Top-K size, seed papers, profile preferences, like or dislike feedback, follow-up queries, selected-paper explanation mode, and provider configuration. Its output is a RecommendationWorkflow with candidate papers, ranked recommendations, extracted briefing items, the final daily briefing, and a workflow trace. Each trace step records status, evidence source, fallback state, input and output summaries, and structured errors, so the user can inspect how the Agent reaches each result.

The user-facing outputs cover three workflows. The Agent generates a daily briefing with an executive summary, summary table, highlighted paper, Top-K paper notes, comparisons, reading priorities, and evidence boundary. It records feedback and returns a refined ranking with rank and score movement. It also supports follow-up filtering over locally stored papers and fetches from arXiv only when local results are empty and the workflow allows retrieval.

2 Agent Design

Figure 1 shows the implemented architecture. The orchestrator coordinates Skills, records workflow traces, and preserves the shared result contract. DiscoveryRecommendationSkill wraps seed parsing, query planning, arXiv retrieval, deterministic ranking, semantic seed ranking, feedback refinement,

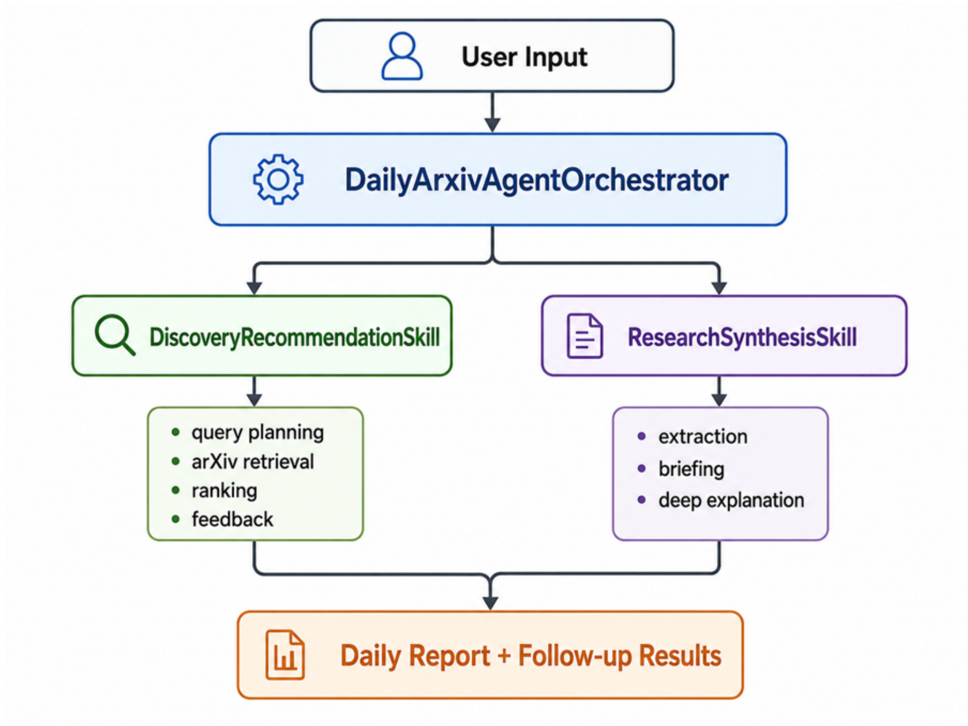

Figure 1: Agent architecture. DailyArxivAgentOrchestrator coordinates two public Skills through structured contracts and a shared SQLite state layer. DiscoveryRecommendationSkill handles query planning, retrieval, ranking, feedback, and follow-up. ResearchSynthesisSkill handles extraction, briefing, and selected-paper explanation.

and follow-up filtering. ResearchSynthesisSkill wraps per-paper extraction, daily briefing generation, and selected-paper deep explanation. The older fine-grained Skills remain available internally, which keeps testing precise while presenting the deliverable as a clean Agent with two public Skill boundaries.

The workflow begins by building a RetrievalQuery. When seed papers are provided, SeedParsingSkill converts arXiv IDs, arXiv URLs, or titles into a SeedPreference. QueryPlanningSkill creates a QueryPlan with required terms, optional terms, phrases, categories, and query variants. Invalid LLM-assisted plans fall back to deterministic planning before retrieval. ArxivRetrievalSkill calls the arXiv Atom API when needed, parses metadata, and caches normalized PaperMetadata records in SQLite.

Ranking has two paths. The deterministic path combines topic matches, query-plan terms, phrases, category and date context, seed preference text, retrieval provenance, and feedback events. The semantic seed path checks embedding readiness, fails closed when configuration is unsafe, and otherwise compares seed and candidate embeddings while retaining lexical, category, and recency signals. Both paths output Recommendation rows with rank, score, rationale, evidence source, and score breakdown.

ResearchSynthesisSkill consumes the Top-K recommendations rather than retrieving or reranking papers. PaperExtractionSkill creates PaperBriefingItem records with summaries, contribution claims, method cues, relevance rationales, and reading guidance. DailyBriefingSkill combines those items with candidate-pool, query-plan, retrieval, and ranking context. PaperDeepExplanationSkill supports method, experiment, and limitations modes. It prefers full text when available, then falls back to abstract or metadata with explicit labels, preventing weak evidence from being presented as full-paper analysis.

SQLitePaperStore stores paper metadata, retrieval runs, seed preferences, feedback events, full-text cache rows, and embedding cache rows. This shared layer keeps the two public Skills independent.

Table 1: Fixture-backed group-level evaluation. These results measure reproducible workflow behavior under controlled inputs, not broad benchmark performance.

Evaluation area	Metric	Result
Search quality	relevant candidate coverage	1.0
Recommendation	precision@3 / recall@3 / MRR	1.0 / 1.0 / 1.0
Ranking explainability	rationale coverage	1.0
Daily briefing	required-section coverage	1.0
Daily briefing	Top-K coverage	1.0
Daily briefing	forbidden full-text claims	none
Deep explanation	method-mode completeness	0.86
Full project tests	pytest result	195 passed

All public methods return SkillResult objects with success, empty, fallback, or error status, so the CLI, UI, tests, and report use the same interpretation of output and failure cases.

3 Related Work

The project builds on scholarly information retrieval, recommender systems, semantic similarity, and evidence-grounded generation. arXiv metadata access follows the arXiv API (arXiv, 2026). The recommendation design follows ranked retrieval practice by reporting precision, recall, and mean reciprocal rank, and by separating candidate generation from ranking (Manning et al., 2008). It also follows hybrid recommender-system ideas, where topic, category, content, seed papers, and user feedback provide complementary signals (Burke, 2002). Semantic seed ranking follows sentence embedding similarity, where related texts are represented in a vector space so that seed papers can personalize ranking beyond exact keyword overlap (Reimers and Gurevych, 2019).

The synthesis side follows retrieval-augmented generation, where generated text is conditioned on retrieved evidence and exposes the boundary between source material and generation (Lewis et al., 2020). Faithfulness research in summarization shows that fluent summaries may contain unsupported claims, so the Agent treats evidence labels and abstentions as part of the output contract (Maynez et al., 2020).

4 Results

Evaluation uses deterministic fixtures and helper metrics in daily_arxiv_agent.evaluation.metrics. The goal is to verify reproducibility, output coverage, measurable ranking and feedback effects, and evidence discipline, not to claim state-of-the-art recommendation quality. The main search-quality fixture contains exact, related, weak, and unrelated candidates for multimodal LLM agents for robotic manipulation.

Table 1 summarizes the main results. Broad planned search reaches relevant candidate coverage of 1.0 and ranks the three expected relevant papers in Top-K, giving precision@3, recall@3, and MRR of 1.0. Rationale coverage of 1.0 means every recommendation carries an inspectable explanation. The briefing evaluator reports required-section and Top-K coverage of 1.0 and rejects forbidden full-text claims. The method-mode explanation score is 0.86 because the fixture intentionally omits one required method field, which the evaluator reports rather than hiding inside generic text.

The qualitative demo has two cases. With a world model topic and seed papers about world action models, recommendations shift toward robotics-oriented world-model papers. With no fresh topic, the stored profile uses saved seeds and feedback to produce recommendations aligned with robotics and deep learning. These cases show that the Agent is more than a stateless arXiv search wrapper: it remembers preference signals and exposes their ranking effects.

5 Analysis

Table 2 makes the main ablations explicit. Query planning improves relevant candidate coverage from 0.33 to 1.0 because it searches beyond a brittle strict phrase. Semantic seed ranking changes

Table 2: Quantitative ablation summary. Each row uses deterministic fixtures or trace counters from the implemented workflow.

Component	Metric	Baseline	Full setting
Retrieval	relevant candidate coverage	0.33 strict phrase	1.0 planned search
Ranking	precision@1	0.0 lexical distractor	1.0 semantic seed
Feedback	moved papers / unchanged papers	0 / 3 before feedback	2 / 1 after feedback
Briefing	claim specificity score	< 0.6 generic summary	≥ 0.6 enhanced briefing
Evidence	forbidden full-text claims	≥ 3 violating claims	0 evidence-bounded claims
Explanation	method-section completeness	incomplete field detected	0.86, with 1 missing field
Follow-up	arXiv requests on local hit	1 optional fetch path	0 local-first path

the top result in the seed-personalization fixture, improving precision@1 from 0.0 to 1.0 when a lexical distractor has high keyword overlap but low semantic similarity. Feedback refinement is measured by rank movement rather than subjective preference: two papers move and one remains unchanged after feedback. Enhanced briefing passes the specificity gate that generic text fails, and the evidence-boundary check reduces forbidden full-text claims to zero. The explanation score remains 0.86 because the evaluator reports one missing method field. Local-first follow-up reduces arXiv calls to zero when stored papers already satisfy the query.

Several limitations remain. The metrics are fixture-backed, so they validate workflow correctness and evidence discipline rather than general relevance quality across arXiv. Live arXiv retrieval and live LLM calls are nondeterministic because results, provider behavior, and latency can change. Fake providers make tests reproducible but do not measure human preference for writing quality. Default briefings do not parse PDFs and should not be read as full-paper surveys. A stronger future evaluation should include human relevance labels, live-provider latency and cost, and longitudinal feedback logs.

Overall, the Agent satisfies the project requirements. It integrates public Skills into a coherent workflow, defines clear inputs and outputs, reports results and ablations, provides visualization, and preserves the central evidence boundary: users can see which claims come from metadata, abstracts, rankings, candidate-pool context, or full text.

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